

Evaluating an Offline Detector Against Adversarial Prompt Attacks

A Reproducible Framework for Prompt-Attack Generation and Defensive Evaluation

Evidence status: Refreshed against current code snapshot 08b40538c00146981d450cad7af5cedd87b887b6 and successful CI run 36783655822 on 2026-09-30.

Contribution

A reproducible offline prompt-attack evaluation framework that reports both the stronger in-distribution result and the weaker novel-phrasing result, making the detector's generalization gap explicit instead of hiding it behind a random split.

Method

1. Generate adversarial and benign prompt corpora across defined attack categories.
2. Train/evaluate the offline TF-IDF + Logistic Regression detector.
3. Compare random, grouped-template, structural-fixture, and novel-phrasing OOD evaluations.
4. Emit JSON/SARIF evidence for CI.
5. Keep live-model behavior claims out of scope unless separately measured.

Current Verified Evidence

- Current-main Python 3.12 CI reports:
- 175 passed, 1 skipped.
 - 94.30% statement coverage; CI gate is 90%.
 - Current benchmark values remain pinned:
 - random split F1 1.00
 - grouped-template F1 0.9714
 - novel-phrasing OOD F1 0.7188
 - novel-phrasing precision 0.5897
 - novel-phrasing recall 0.92
 - novel-phrasing false-positive rate 64% (16/25)
 - Lint, type checking, dependency audit, performance gate, external validation, and comparable OOD benchmark jobs completed successfully.

Reproducibility

Expected CI evidence: 175 passed, 1 skipped, 94.30% coverage.

Honest Boundaries

- Offline detector evaluation does not prove whether a live LLM would follow an injected instruction.
- English/template coverage is finite.
- Structural fixtures are not equivalent to true OOD natural-language paraphrases.
- No real-world jailbreak-defense rate is claimed.